

Of course, after we have decided what postulates to use, we must prove our theorems by using *these postulates and not by looking at other figures*.

Postulate Group I. This group contains three postulates, indicated by I-1, I-2, and I-3.

I-1 *The three-point postulate.* There are at least three points which do not all lie on one line.

I-2 *The point-line postulate.* For any two different points there is exactly one line containing these points.

I-3 *The line-point postulate.* Every line contains at least two points.

The point-line postulate tells us one way we want lines to behave. The line-point postulate *starts* us on the way to getting points on a line. The three-point postulate tells us that not all points are on one line. It gets us off the line.

With only these postulates we cannot as yet prove very much. The following two theorems are about all that can be formulated.

Theorem 3-1. *Two different lines do not intersect in more than one point.*

Proof. The theorem is either true or false. We shall prove that it cannot be false. Hence it must be true. We therefore suppose that the lines intersect in at least two points and then show that this is impossible. Suppose the lines are l and m and that they intersect in the two points A and B (Fig. 3-6). We would then have two different lines, l and m , through A and B . This contradicts the point-line postulate. Therefore our assumption that l and m intersect in more than one point is false and our theorem is true.

Remark. Several ideas from logic are involved in this proof. Comment on them.

Theorem 3-2. *There are at least three lines in the plane.*



FIGURE 3-6

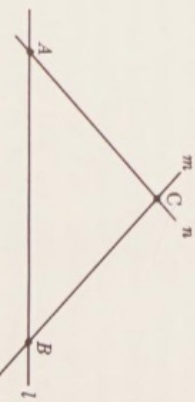


FIGURE 3-7

Proof.

STATEMENTS

1. There are at least three points in the plane not all on one line. Call them A , B , and C .
2. Let l be the line through A and B , m the line through B and C , and n the line through C and A .
3. The three lines l , m , and n are all different from one another.

REASONS

1. Three-point postulate.
2. Point-line postulate.
3. If any two were the same line, all three points A , B , and C would be on one line, contrary to the way they were chosen.

The three points and lines are shown in Fig. 3-7. Of course, it looks just like the drawing (Fig. 3-5) that suggested our first postulates.

Remark. At this point we cannot prove from our postulates that there are any more points or lines than shown in Fig. 3-7.

Exercise Group 3-1

1. Restate the point-line and the three-point postulates in other words.
2. Prove that if l is any line, there is at least one point not on l .
3. Explain why both a and b in Fig. 3-8 cannot represent lines if P and Q represent points.



FIGURE 3-8

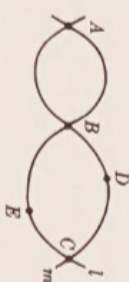


FIGURE 3-9

9. A line l contains the points P and Q . The line m also contains P and Q . What can you conclude?
10. Two lines have a point A in common. What can you say about the lines?
11. Prove that there are at least two lines on each point.
12. Why, when looking at Fig. 3-10, can we not prove that there are at least five points in the plane?

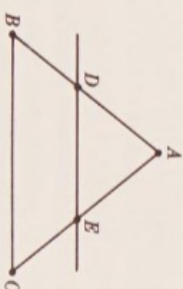


FIGURE 3-10

4. Prove that if l is a line and P is on l then there is another point on l .
5. Why can we not prove there are more than three lines in our plane?
6. State what "intersect" means.
7. Suppose that l is a line. List all the things you can say about this line and points of the plane. Give your reasons.
8. Given Fig. 3-9, what can you conclude about l and m ? Why?

are never between any others. These are called the *endpoints*. The other two points are between points exactly twice.

The line-building postulate tells us first that there is always a point C "beyond" B from A . In fact, Euclid stated (vaguely) something very much like this as a postulate. But our postulate also tells us that there are points between any two. Euclid never stated this. It is this postulate that gives the existence of more than two points on a line, and therefore it is called the line-building postulate. You might think about the relation between this postulate and the real world. When you draw pictures of points, can you always mark a picture of a third point between any two?

The two-side postulate says that a line has two sides. It merely expresses the way we tell whether or not two of us are on the same or on different sides of a fence. If, when we look at each other, we see the fence between us, we know we are on opposite sides of it. If we do not see the fence between us, we are on the same side.

Now, using these postulates, we can define some familiar geometric objects.

Definition 3-2. If A and B are any two points, then the set of points consisting of A and B and all the points between A and B will be called the *line segment* (or *segment*) determined by A and B . The segment will be indicated by " AB ." The points A and B are called the *ends* of the segment. In Fig. 3-16, P is a point of the segment AB .



FIGURE 3-16



FIGURE 3-17

Definition 3-3. If A , B , and C are three points not all on one line,* the set of points of the segments AB , BC , AC is called a *triangle* (Fig. 3-17). We call the points A , B , and C *vertices* of the triangle and the segments AB , AC , BC *sides* of the triangle. To indicate "triangle ABC ," we write " $\triangle ABC$."

Before working the exercises, read all the postulates again to be sure that you understand them.

* Points all lying on a line are said to be *collinear*. Hence in the definition we could have said, "If A , B , and C are not collinear . . ."

Exercise Group 3-2

1. Draw a line and mark five points A, B, C, D, E on it. Write out all the betweenness relations.
2. If ABC and ACD , prove that the four points A, B, C , and D lie on one line.
3. Give in your own words the definitions of "segment," "triangle," and "two points are on the same side of a line."
4. If A, B, C , and D are four points of a line and we know that CAD and ABD , what are the other betweenness relations?
5. If A, B, C , and D are four points of a line and we know that CAD and CAB , what are the other betweenness relations?
6. If E, F, G , and H are four points on a line and we know that EFH , how many possible positions are there for G that can be described in terms of our betweenness relations?

7. A and B are points on a line l . Write three true statements about points either on l or not on l . Give reasons, as below.

STATEMENTS

1. There is a point C such that ACB .
2. _____
3. _____
4. _____

REASONS

1. The line-building postulate.
2. _____
3. _____
4. _____

8. If RST and VKS , relabel the points R, S, T , and V as A_1, A_2, A_3 , and A_4 (in some order) in accordance with II-3. If CAD, BDE , and BCA , relabel the points A_1, A_2, A_3, A_4 , and A_5 in accordance with II-3.

9. Prove that there are at least four lines on each point.

10. If DEF and EGF , then the four-point betweenness postulate implies that _____ and _____.

11. If RST , then what postulate tells us that R, S , and T are distinct points on a line?

*12. Draw a line and choose two points on it, A and B (Postulate I-3). Mark a point C such that ABC (Postulate II-4). Mark a point D such that BCD (Postulate II-4). Is it true that ACD ? Why? Use II-3 to prove that ACD .

*13. Draw a line and choose two points on it, A and B . Choose a point C such that ACB (Postulate II-4). Now choose a point D such that CDB . Is it true that ACD ? that ADB ? Use II-3 to prove this.

Now choose a point E such that DEB . What are all the betweenness relations between these five points? Can this process be indefinitely repeated? Describe, in your own words, the meaning of "this process." How many points are there on a line segment?

*14. Draw a line and choose a point A on one side and a point B on the other side. Now mark a point C on the opposite side of the line from A . Points B and C will be on the same side of the line. How does this follow from II-5?



FIGURE 3-18

*15. A line l (Fig. 3-18) does not contain any of the vertices of a triangle. The line l does intersect the side AB .

3-5 RAYS AND ANGLES

It is a familiar fact from experience that a point O on a line l divides the other points of the line into two sets—those on one side and those on the other side of the point O on the line l . (See Fig. 3-19.) From our postulates on betweenness we can now make precise definitions of this observation. First we need to know that a point on a line separates the line.

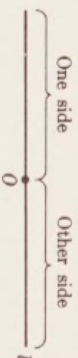


FIGURE 3-19

Theorem 3-3 The two-side theorem. Let O be any point of a line l . The points of l , different from O , are separated into two sets such that:

- (1) If P and Q are in the same set, then either OPQ or OQP .
- (2) If P is in one set and Q is in the other set, then POQ .

**Proof.* Choose a point A not on l and consider the line m through O and A (see Fig. 3-20). We are using I-1 and I-2. By the two-side postulate, II-5, the line m separates the plane into the two sides of m . Let one of the sets, which we seek, consist of the points of l which are on one side of m , and let the other set consist of the points of l on the other side of m .

If P and Q are in one of these sets, they are by definition on the same side of m , and therefore O is not between them. Hence by II-2 we must have either OPQ or OQP . This proves (1).

Now suppose that P is in one set and Q in the other set (see Fig. 3-21). Then by definition of the sets, P and Q are on different sides of m . By

but does not intersect the side BC . Prove that l must intersect AC .

*16. Draw a line and choose two points A and B on one side of the line. Choose C so that ACB . Can you prove that C and A are on the same side of the line?

*17. If A is on line l and B is not on l and ACB , prove that C and B are on the same side of l .

*18. Prove that for every line l there is a point on each side of l .

* Starred proofs are difficult and may be omitted. However, the meaning of the theorems must be understood.

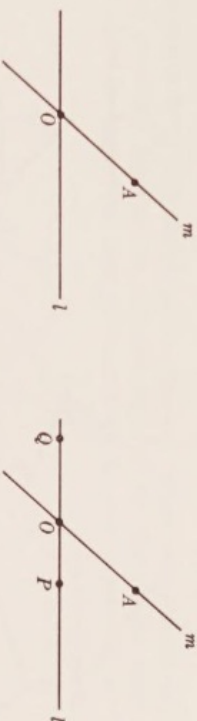


FIGURE 3-20

the two-side postulate, II-5, there is a point of m between P and Q . But l and m have only point O in common. Hence the point between must be O , and we have POQ . This proves (2).

Remark. The two sets of the theorem are not dependent on how we choose the line m , because we end up with the same betweenness relations.

Definition 3-4. The two sets of Theorem 3-3 are called the two *sides* of O on the line l .

Definition 3-5. A ray from O , or a ray with end O , is a set of points consisting of O and all the points on one side of O of a line l through O (see Fig. 3-22).

If P is any point of the ray, we may speak of "ray OP ." Note that we may describe the ray as the set of points consisting of O and P and all points Q such that OQP and all points R such that OPR . Refer back to Theorem 3-3.



FIGURE 3-22

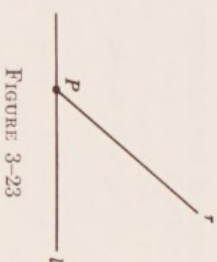


FIGURE 3-23

Exercise 3-3

Consider a line l and a ray r , not on l , from a point P on l , as in Fig. 3-23. What do we mean by " r 's side of l "?

Definition 3-6. The set of all points on two rays from a point is called an *angle*. The point is called the *vertex* of the angle, and the two rays are called the *sides* of the angle. If the two rays are on the same line, the angle is called a *straight angle*. If O is the vertex and A and B are points other than O on the sides of the angle, we shall speak of "angle AOB " and write this as " $\angle AOB$." (See Fig. 3-24.)